DRISHTI ([redacted])

Explainable AI-Assisted Diabetic Retinopathy Screening & Decision Support System

Smart India Hackathon 2026 | PS-26038 | MathWorks

1. Executive Summary & Problem Breakdown

Drishti addresses India's diabetic blindness crisis by deploying an explainable, multi-factor safety-gated AI screening platform directly to rural Primary Health Centres (PHCs). With over 77 million diabetic patients and fewer than 20,000 ophthalmologists nationwide, automated triage of routine cases is mathematically essential to eliminate clinical backlog and prevent irreversible vision loss.

Module / Capability	Implementation Status	Verified Test Metric / Evidence
Image Quality Gating	IMPLEMENTED	Laplacian Variance + Illumination + FOV Score >= 0.70
5-Class DR Classification	IMPLEMENTED	ResNet-18 | 76.87% Accuracy | QWK = 0.870
Referable DR Triage	IMPLEMENTED	Specificity = 96.62% | Sensitivity = 82.14% | AUC = 0.980
Explainable AI (Grad-CAM)	IMPLEMENTED	layer4[1].conv2 Backpropagation Attention Overlays
Human-in-the-Loop Telemedicine	IMPLEMENTED	Review Queue + Validate/Override Actions + Audit Trail
District Telemedicine Scaling	IMPLEMENTED	MATLAB/Simulink M/M/c Model (120,000 Annual Patients)
Offline Rural Sync Queue	IMPLEMENTED	SHA-256 Hash Idempotent SQLite / Storage Cache

2. Deep Learning Validation on Real APTOS 2019 (Held-Out Test Set)

The model was evaluated strictly on a held-out test split of 549 images with zero data leakage. Multi-class Quadratic Weighted Kappa reached **0.870**, reflecting substantial agreement with international clinical standards. Binary referable specificity achieved **96.62%** with an ROC AUC of **0.980**. Measured sensitivity was **82.14%**, with errors concentrated in adjacent Grade 1 vs Grade 2 distinctions safely mitigated by mandatory clinician review.

Ground Truth	Pred L0	Pred L1	Pred L2	Pred L3	Pred L4	Class F1
Level 0 (No DR)	256	12	2	0	0	96.8%
Level 1 (Mild NPDR)	3	43	9	0	0	57.3%
Level 2 (Moderate NPDR)	0	36	92	13	9	66.2%
Level 3 (Severe NPDR)	0	0	12	9	8	31.6%
Level 4 (Proliferative DR)	0	4	13	6	22	52.4%

3. District Telemedicine Capacity Simulation (MATLAB/Simulink)

Simulated 120,000 annual diabetic screenings across 300 working days (arrival rate lambda = 50 patients/hr). In the baseline manual system with 2 ophthalmologists (capacity mu = 48/hr), doctor utilization hits **104.2%**, causing infinite queue backlog and >48-hour delays. Drishti's automated AI triage clears ~72% of healthy patients, reducing specialist arrival rate to lambda = 14/hr (utilization **29.2%**) and cutting review delay to under **4.2 minutes**.